12TH CHEMISTRY · CHAPTER 26 · FULL CHAPTER NOTES

Polymers

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What are polymers and monomers? Why are organic polymers important?

Polymers are very large molecules made by joining many small molecules called **monomers** through polymerisation. Organic polymers are modern materials of great significance: they have made our lives easier, more efficient and highly economical due to their superior properties and affordable costs. They are seen in daily life in shopping bags (poly(ethene)), clothesline (poly(propene)), bulletproof vests (Kevlar) and aerospace applications.

Q.2 [Short] On what do the properties of polymers depend? Give examples of soft and hard polymers.

The properties of polymers vary broadly depending on the type and number of monomers in the polymer, and on the method by which they are manufactured. Some polymers are very light and soft, and are suitable where soft materials are needed, e.g. poly(ethene) and poly(propene). Others are hard and thermally very stable, and are used as hard and durable materials, e.g. Kevlar, which is suitable for bulletproof vests.

Q.3 [Short] How are synthetic polymers named systematically?

Synthetic polymers are named systematically by writing the prefix “poly” before the name of the monomer in brackets. For example, polythene is poly(ethene) and polyvinyl chloride is poly(chloroethene). For polymers with two or more monomers, common names are usually used because the systematic name is too long: poly(ethane-1,2-diyl benzene-1,4-dicarboxylate) is commonly called polyethylene terephthalate, PETE.

26.1 POLYMERISATION

Q.4 [Short] What is polymerisation? Name its two main types.

Polymerisation is the process of joining monomers to form a polymer. There are two main types: (i) addition polymerisation and (ii) condensation polymerisation.

Q.5 [Long] What is addition polymerisation? What structural feature must the monomer have?

In addition polymerisation, monomers add to each other one by one, forming long chains. All the atoms from the original monomers become part of the polymer; nothing is lost.

The monomers must have a carbon–carbon double (C=C) or triple (C≡C) bond. This bond “opens up” to allow the monomers to link together. No small molecules (like

water) are produced as side products. It often happens through a chain reaction started by a very reactive molecule (an initiator, which forms free radicals).

Q.6 [Short] How is poly(ethene) formed from ethene?

A large number of ethene ($CH_2=CH_2$) molecules add together by breaking their double bonds and linking end-to-end; n is very large, e.g. up to 10 000.

$$n\ CH_2=CH_2 \rightarrow -[CH_2-CH_2]-n \text{ (initiator, heat/pressure)}$$

Q.7 [Long] Differentiate between LDPE and HDPE.

Low density poly(ethene), LDPE	High density poly(ethene), HDPE
Its chains are more branched.	Its chains are linear.
The chains cannot pack tightly.	The chains pack closely to give a compact structure.
It is a less solid, more elastic, very light and soft material.	It is a tough and dense material.
Used for shopping bags, light plastic bottles, etc.	Used for plastic containers, thick shopping bags, toys, etc.
—	Made from ethene using a Ziegler–Natta catalyst at low pressure and temperature.

Q.8 [Short] Why is LDPE softer and more elastic than HDPE?

LDPE chains are more branched, so they cannot pack tightly; this gives a less solid and more elastic material. HDPE chains are linear and pack closely to give a compact structure, so the material is tough and dense.

Q.9 [Short] What are Ziegler–Natta catalysts?

Ziegler–Natta catalysts are a Nobel Prize-winning invention that revolutionised plastic production. The catalyst was initially discovered by Karl Ziegler in 1953. It enabled, for the first time, the low-pressure polymerisation of ethene into linear, high-density poly(ethene) (HDPE). The same catalyst family was expanded by Giulio Natta to create stereoregular polymers, such as crystalline polypropylene.

Q.10 [Short] How is PVC prepared?

Poly(chloroethene) or PVC is made from the monomer chloroethene ($CH_2=CHCl$) through an addition polymerisation reaction.

$$n\ CH_2=CHCl \rightarrow -[CH_2-CHCl]-n \text{ (initiator)}$$

Q.11 [Short] What are plasticizers? How do they work?

PVC is a versatile plastic whose properties can be changed a lot by adding different chemicals (plasticizers). **Plasticizers** are chemicals added to materials to make them softer, more flexible and easier to work with. They do this by reducing the strong forces between the polymer chains, allowing the chains to move more freely.

Q.12 [Long] Compare rigid PVC with plasticised PVC.

Rigid PVC (without plasticizers)	Plasticised PVC (with plasticizers)
Hard and stiff; does not bend easily.	Flexible and soft; can be bent and shaped easily.
Chemically inert (does not react with most acids, bases or oils); a good electrical insulator.	Keeps good chemical and electrical properties, but is less rigid.
Because of its chlorine atoms it is naturally resistant to burning and self-extinguishing; it lasts long and resists weathering.	Strong forces between chains are reduced, so the chains move more freely.

Q.13 [Short] Why is rigid PVC resistant to burning?

Due to the chlorine atoms in its chains, PVC is naturally resistant to burning and is self-extinguishing.

Q.14 [Long] Give the industrial applications of PVC.

1. Pipes and fittings for water supply, drainage and sewage systems, due to its corrosion resistance and durability.
2. Window frames and door profiles, floor vinyl tiles and sheets.
3. Electrical insulation for wires and cables.
4. Tubing, blood bags and catheters.
5. Cling films, bottles (e.g. for cooking oil), blister packs.
6. Raincoats, artificial leather, inflatable products (e.g. swimming pools).

Q.15 [Long] What is condensation polymerisation? What are the requirements for the monomers?

In condensation polymerisation, monomers join together, but each time a new bond forms, a small molecule is eliminated from the monomers. This small molecule is usually water; it can also be HCl, CH_3OH or NH_3.

The monomers must have at least two reactive functional groups (like –OH, –COOH, $–NH_2$) that can react with each other; these groups are on different ends of the monomers. Small molecules are always produced as a side product. The reaction typically happens in a “step-by-step” manner.

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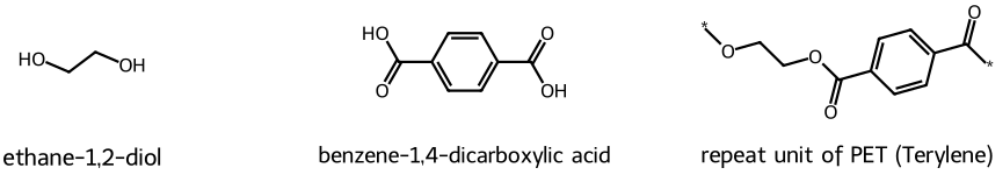
Q.16 [Long] Compare addition and condensation polymerisation.

Addition polymerisation	Condensation polymerisation
Monomers add to each other one by one; all atoms of the monomers become part of the polymer.	Each time a new bond forms, a small molecule is eliminated from the monomers.
Monomers must have a C=C double (or C≡C triple) bond.	Monomers must have at least two reactive functional groups (–OH, –COOH, $–NH_2$, –COCl) on different ends.
No small molecule is produced as a side product.	A small molecule (H_2O, HCl, CH_3OH or NH_3) is always produced.
Often a chain reaction started by an initiator (free radicals).	Typically takes place in a step-by-step manner.
The main chain consists only of carbon atoms.	The main chain contains O or N atoms with carbon (ester or amide links).
Examples: poly(ethene), PVC, poly(propene), PTFE.	Examples: polyesters (Terylene/PET) and polyamides (nylon-6,6, Kevlar).

Q.17 [Long] What are polyesters? How is Terylene (PET) prepared?

Polyesters have **ester links** (–CO–O–) in their backbone. A diol (a molecule with two –OH groups) and a dicarboxylic acid (a molecule with two –COOH groups) are combined to give a polyester. The –OH of the diol reacts with the –COOH of the acid and a molecule of water is removed. Example: Terylene (PET) from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

$$n\ HO–CH_2CH_2–OH + n\ HOOC–C_6H_4–COOH \rightarrow$$
$$–[O–CH_2CH_2–O–CO–C_6H_4–CO]–n + 2n\ H_2O$$



ethane-1,2-diol benzene-1,4-dicarboxylic acid repeat unit of PET (Terylene)

Monomers and repeat unit of PET (shows the bond to the next unit)*

Q.18 [Short] Give the properties and uses of polyesters.

Polyesters have special features, such as high tensile strength, which make them suitable for manufacturing fibres. Polyester is used as an alternative to cotton due to its low cost and ability to resist creases. It is also used as a plastic: PETE is used to make plastic bottles for beverages and other food packaging. Polyester is also used as cross links to obtain hard plastics, e.g. glass fibre reinforced cross-linked polyester.

Q.19 [Long] What are polyamides? Describe three ways in which polyamides are formed.

Polyamides have **amide links** (–CO–NH–) in their backbone. They can be prepared from many starting materials.

i) From a diamine and a dicarboxylic acid

A diamine (a molecule with two $–NH_2$ groups) and a dicarboxylic acid (two –COOH groups) are joined together to give a polyamide. The $–NH_2$ of the diamine reacts with the –COOH of the acid and a molecule of water is removed. Example: nylon-6,6 uses hexamethylenediamine and adipic acid.

$$n\ H_2N–R–NH_2 + n\ HOOC–R'–COOH \rightarrow –[HN–R–NH–CO–R'–CO]–n + 2n\ H_2O$$

ii) From a diamine and a dioyl chloride

A diamine and a dioyl chloride (a molecule with two –COCl groups) are joined to give a polyamide. Dioyl chlorides are more reactive than dicarboxylic acids. The $–NH_2$ of the diamine reacts with the –COCl of the dioyl chloride and a molecule of hydrogen chloride (HCl) is removed.

$$n\ H_2N–R–NH_2 + n\ ClOC–R'–COCl \rightarrow –[HN–R–NH–CO–R'–CO]–n + 2n\ HCl$$

iii) From an amino acid

An amino acid is a single monomer that has both an $–NH_2$ (amino) group and a –COOH (carboxylic acid) group in one molecule. The $–NH_2$ group of one amino acid reacts with the –COOH group of another, removing water to form a peptide bond, which is also a type of amide linkage. Proteins are natural polyamides made from amino acids.

$$n\ H_2N–CH_2–COOH \rightarrow –[HN–CH_2–CO]–n + n\ H_2O$$

The general equations are written here with R and R′ and with balanced amounts of the small molecule (2n). The textbook prints them with $–CH_2–$ groups and "nH_2O / nHCl".

Q.20 [Short] Why is a dioyl chloride sometimes used in place of a dicarboxylic acid?

Dioyl chlorides are more reactive than dicarboxylic acids, so the polyamide forms more easily; HCl is eliminated instead of water.

Q.21 [Long] Describe the condensation reaction of ammonia or an amine with an acyl chloride.

An amide linkage is also formed between an acyl chloride (a very reactive organic compound) and ammonia or an amine, e.g. methanamine ($CH_3–NH_2$) or N,N-dimethylamine, $(CH_3)_2NH$. A molecule of hydrogen chloride (HCl) gas is removed. The reaction happens easily at room temperature because acyl chlorides are fairly reactive.

$$R–COCl + R'–NH_2 \rightarrow R–CO–NH–R' + HCl$$

$$CH_3–COCl + NH_3 \rightarrow CH_3–CO–NH_2 + HCl$$

26.2 NYLON

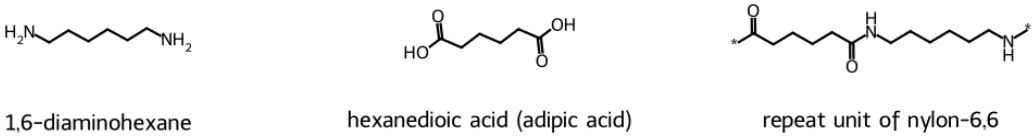
Q.22 [Short] What is nylon?

Nylon is a type of synthetic polyamide. It is formed by condensation polymerisation, usually from two different monomers, a diamine ($H_2N–R–NH_2$) and a dicarboxylic acid (HOOC–R–COOH). During the reaction a small molecule, typically water, is removed.

Q.23 [Long] How is nylon-6,6 prepared?

Nylon-6,6 is the most common example of synthetic amides. It is formed by a polymerisation reaction between 1,6-diaminohexane and hexanedioic acid (adipic acid); each monomer has six carbon atoms.

$$n\ H_2N–(CH_2)_6–NH_2 + n\ HOOC–(CH_2)_4–COOH \rightarrow$$
$$–[HN–(CH_2)_6–NH–CO–(CH_2)_4–CO]–n + 2n\ H_2O$$



Monomers and repeat unit of nylon-6,6 (shows the bond to the next unit)*

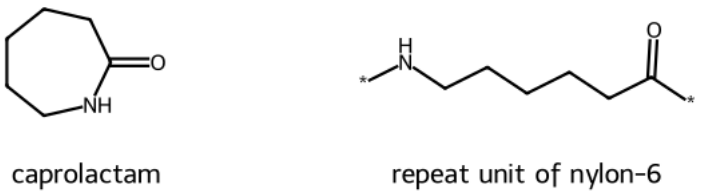
The textbook names the acid "1,4-hexanedioic acid"; the six-carbon diacid drawn in its figure, $HOOC–(CH_2)_4–COOH$, is hexanedioic acid (adipic acid).

Q.24 [Short] Why are skateboard wheels made from nylon?

Because of its hard, smooth and shock-free nature.

Q.25 [Long] What is nylon-6? Why is it classified as a condensation polymer?

Nylon-6 is formed from a single monomer called **caprolactam**, which has a cyclic structure. It is named nylon-6 because the monomer has six carbon atoms. The reaction is not a typical condensation reaction because no small molecule is removed. Nevertheless, nylon-6 is classified as a condensation polymer due to the mode of formation of the amide linkage.



Caprolactam and the repeat unit of nylon-6 (shows the bond to the next unit)*

Q.26 [Short] Compare the properties of nylon-6 with nylon-6,6.

Nylon-6 is a harder material than nylon-6,6 and provides a smoother surface and higher impact resistance (harder to break by hitting). It is easier to mould because its melting point is lower than that of nylon-6,6. It is mostly used in rollers, skateboard wheels and power tool handles to avoid shocks.

Quick Check 26.1

Q.27 [Short] a) The repeating unit of nylon-6, 10 is shown below. Draw the structures of its monomers from this structure. –[NH–$(CH_2)_6$–NH–CO–$(CH_2)_8$–CO]–

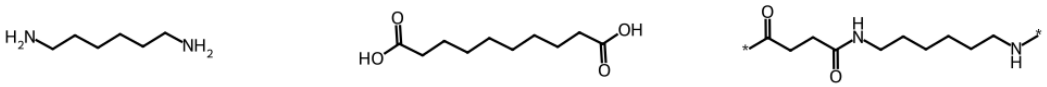
Break the amide bonds; add –H to each N and –OH to each C=O. The monomers are **hexane-1,6-diamine** (6 carbon atoms) and **decanedioic acid** (10 carbon atoms).

$H_2N–(CH_2)_6–NH_2$ and $HOOC–(CH_2)_8–COOH$

Q.28 [Short] b) Monomers of nylon-6,4 are given below. Draw the structure of the repeating unit of the polymer formed from these monomers. $H_2N(CH_2)_6NH_2$ $HOOC(CH_2)_2COOH$

An amide link forms between each amino group and each carboxylic group, with loss of water. The repeating unit is:

–[HN–$(CH_2)_6$–NH–CO–$(CH_2)_2$–CO]–n



hexane-1,6-diamine decanedioic acid repeat unit of nylon-6,4

Monomers of nylon-6,10 and the repeat unit of nylon-6,4 (shows the bond to the next unit)*

Q.29 [Short] State the properties of nylon.

- Strong and resistant to breaking; can withstand rubbing and wear and tear.
- Stretchable due to elasticity.
- Slippery surface, which reduces friction.
- Generally resistant to oils, many solvents and alkalis.
- High melting point.

Q.30 [Long] Give the industrial applications of nylon.

- Clothing (stockings, activewear, swimwear), carpets and ropes, due to its strength and elasticity; its fabric is used for raincoats, sportswear and lingerie.
- Mechanical parts of the automotive industry such as gears, bearings, bushings, engine covers and fuel lines, due to its strength, low friction and heat resistance.
- Machine parts such as screws, nuts and bolts, and industrial components where strength and wear resistance are needed.

- Ropes, fishing nets and lines, due to its high tensile strength and resistance to water wear.
- Consumer goods like zippers, toothbrushes, hairbrushes and various moulded plastic parts.

26.3 IDENTIFYING A POLYMER FROM ITS SECTION

Q.31 [Long] How is the type of a polymer identified from a section of its structure?

To identify a polymer from a skeletal or structural formula, look for the repeating linkage and the monomer units.

1. Identifying the type of polymer: first check which atoms make up the main chain (the backbone).

- If the main chain consists only of carbon atoms (–C–C–C–C–), it is an **addition polymer**; these are formed from alkenes.
- If the main chain contains oxygen (O) or nitrogen (N) atoms with carbon, it is a **condensation polymer** (polyester or polyamide).
- If a carbonyl group (C=O) is directly attached next to an oxygen, it is an ester linkage and the polymer is a **polyester**, e.g. PET.
- If a carbonyl group (C=O) is directly next to a nitrogen with a hydrogen attached, it is an amide (peptide) linkage and the polymer is a **polyamide**, e.g. nylon-6,6 or Kevlar.

Functional groups in the polymer	Type	Example
Only C–C bonds in the chain	Addition polymer	Poly(ethene), PVC, PTFE (poly(tetrafluoroethene))
–COO– links	Polyester	Terylene, PLA (poly(lactic acid))
–CONH– links	Polyamide	Nylon-6,6, Kevlar
Benzene rings in the chain	Aromatic polymer	PET or Kevlar

Q.32 [Short] How is the repeat unit of a polymer determined?

2. Determining the repeat unit: identify the smallest section that repeats itself in the section of the polymer.

- Look for where the sequence starts to duplicate itself; locate such duplicating sections in the backbone.
- Place brackets around one of these recurring sections and add the subscript n. The section within the brackets is the repeat unit of the polymer.

Q.33 [Long] How are the monomers identified from the repeat unit of a polymer?

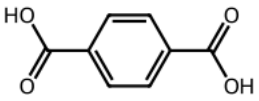
3. Identifying the monomer(s):

- **Addition polymers:** change the C–C single bond in the repeat unit back into a C=C double bond to find the monomer.
- **Condensation polymers:** add water back in. Break the ester or amide bond; add –OH to the C=O side and –H to the O or N side.

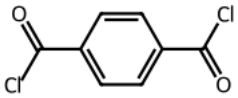
Example: a section of Kevlar gives the repeat unit shown below; breaking its amide bonds gives benzene-1,4-diamine and benzene-1,4-dicarboxylic acid (or its dioyl chloride) as the monomers.



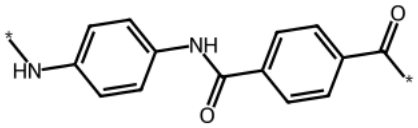
benzene-1,4-diamine



benzene-1,4-dicarboxylic acid



benzene-1,4-dicarbonyl chloride



repeat unit of Kevlar

Monomers and repeat unit of Kevlar (shows the bond to the next unit)*

Quick Check 26.2

Q.34 [Short] **a) Identify the type of polymerisation and monomers forming the following polymer sections.**
i) –CH_2–NH–CO–CH_2–NH–CO–CH_2– ii) a carbon chain with a benzene ring on every second carbon iii) –$CH_2CH(CH_3)$–$CH_2CH(CH_3)$–$CH_2CH(CH_3)$– (polypropylene)

i) The chain contains amide (–CO–NH–) links, so it is a polyamide formed by **condensation** polymerisation. Monomer: aminoethanoic acid (glycine), $H_2N–CH_2–COOH$.

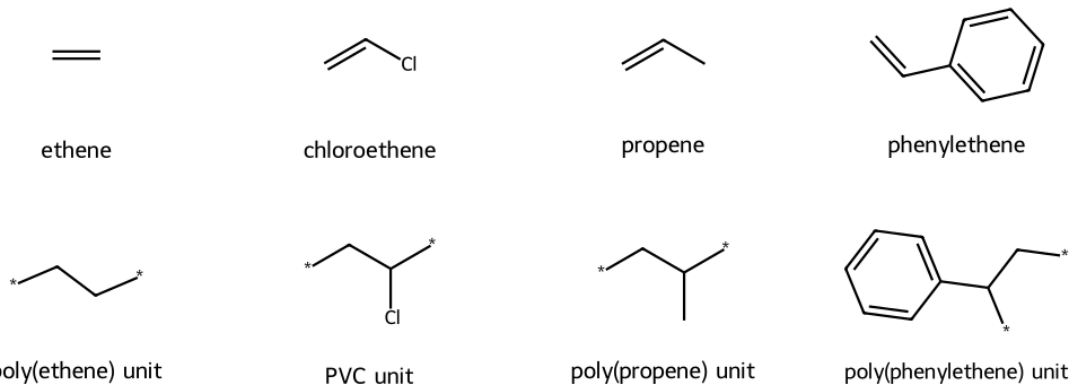
ii) The main chain has only carbon atoms, so it is an **addition** polymer (poly(phenylethene), polystyrene). Monomer: phenylethene (styrene), $C_6H_5–CH=CH_2$.

iii) The main chain has only carbon atoms, so it is an **addition** polymer. Monomer: propene, $CH_3–CH=CH_2$.

26.4 PREDICTING THE TYPE OF POLYMERISATION

Q.35 [Long] Which monomers give addition polymers? What is a copolymer?

When the given monomers contain C=C bonds, addition polymerisation will take place. Examples include ethene, propene and phenylethene (styrene). Mostly homopolymers are formed, but sometimes copolymerisation of two different monomers having double bonds can also be achieved, e.g. a copolymer of ethene and propenoic acid (H_2C=CH–COOH).



Alkene monomers and the repeat units of their addition polymers (shows the bond to the next unit)*

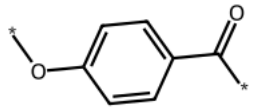
Q.36 [Long] Which monomers give condensation polymers? Which pairs of functional groups can react?

When two functional groups are available to form a bond or bridging linkage to create the polymeric chains, condensation polymerisation takes place. Mostly, a small molecule is removed from the two monomers. The two functional groups may exist in a single molecule (e.g. lactic acid) or in two different molecules (as in nylon-6,6).

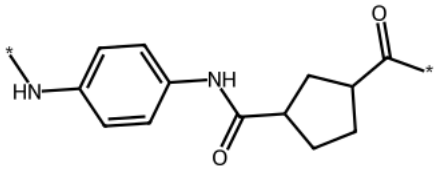
Functional groups	Polymer	Small molecule
amino (–NH_2) + carboxylic (–COOH)	polyamide	H_2O
amino (–NH_2) + acyl chloride (–COCl)	polyamide	HCl
carboxylic (–COOH) + alcohol (–OH)	polyester	H_2O
acyl chloride (–COCl) + alcohol (–OH)	polyester	HCl

Quick Check 26.2 (second box)

Q.37 [Short] **Give the structures of the polymers formed from the monomers. For each polymer identify the following: i. the repeat unit ii. the type of linkage present iii. the attractive force between the polymer chains. a) HO–C_6H_4–COOH (4-hydroxybenzoic acid) b) H_2N–C_6H_4–NH_2 + cyclopentane-1,3-dicarboxylic acid**



(a) repeat unit: polyester



(b) repeat unit: polyamide

Repeat units (shows the bond to the next unit)*

	Polymer (a)	Polymer (b)
i. Repeat unit	–[O–C_6H_4–CO]–	–[HN–C_6H_4–NH–CO–C_5H_8–CO]–
ii. Linkage	Ester linkage (–CO–O–); a polyester	Amide linkage (–CO–NH–); a polyamide
iii. Force between chains	Permanent dipole–dipole forces between the polar ester groups (with instantaneous dipole forces)	Hydrogen bonding between the N–H of one chain and the C=O of another

The forces between polymer chains are not discussed in this chapter; part iii is answered from the intermolecular forces studied earlier.

26.5 CHALLENGE OF NON-DEGRADABLE POLYMERS

Q.38 [Short] **What are non-degradable polymers? Why are they a nuisance?**

Plastic, such as poly(ethene), is the most widely used polymeric material on the globe. Conventional polymers are non-degradable, which means they cannot be broken down into simpler, non-toxic compounds. For this reason they become an environmental nuisance after their disposal: they become part of water bodies, soil and the natural habitats of living organisms. The plastic material is either buried underground or even disposed of in open air where the municipality system is not good.

*The textbook sentence reads “non-degradable, which means they can be broken down…”; the meaning is that they **cannot** be broken down.*

Q.39 [Long] **What challenges are associated with the disposal of non-degradable polymers?**

- **Persistent pollution:** they stay in the environment for a very long time, so they accumulate in landfills, oceans and natural landscapes.
- **Landfill space:** they take up huge amounts of space in landfills because they do not decompose; as population grows, finding space for more landfills becomes difficult.

- **Harm to animals:** animals (especially marine life) can mistake plastic waste for food, leading to choking, internal injuries or starvation; they can also get tangled in plastic debris.
- **Microplastics:** large pieces break down into tiny pieces called microplastics, which spread everywhere, polluting soil, water and even the air; they can enter the food chain and potentially harm human health.

This leads to the need for two major classes of polymers: biodegradable and photodegradable polymers.

Q.40 [Short] What are microplastics?

Over time, large pieces of non-degradable plastics break down into tiny pieces called microplastics. They spread everywhere, polluting soil, water and even the air, can enter the food chain and may harm human health.

Q.41 [Short] Why are poly(alkenes) difficult to biodegrade?

Poly(alkenes) have only strong, non-polar C–C and C–H bonds in their chains and no ester or amide links. They are therefore chemically inert: they are not attacked by acids, alkalis or water (no hydrolysis), and soil organisms cannot break them down. This is why they are difficult to biodegrade.

The textbook lists this as a learning outcome; the explanation in terms of strong C–C and C–H bonds follows exercise MCQ X.

Q.42 [Long] What are biodegradable polymers? Why are polyesters and polyamides biodegradable?

The plastic that is naturally decomposed by microorganisms into simpler non-toxic compounds when buried for a certain period is called biodegradable plastic. Plastics that contain natural polymers, such as starch, are degraded faster, because bacteria in the soil can attack natural polymers easily.

Polyesters and polyamides are more biodegradable than poly(alkene)s because soil organisms can attack them better through **hydrolysis** into simpler components. Polyamides produce carboxylic acids and amines, whereas polyesters are decomposed into carboxylic acids and alcohols. For example, poly(lactic acid) (PLA) can enhance the biodegradability of a plastic.

Q.43 [Short] What are the products of acidic and alkaline hydrolysis of polyesters and polyamides?

Polymer	Acidic hydrolysis gives	Alkaline hydrolysis gives
Polyester (–COO– link)	carboxylic acid + alcohol	carboxylate salt + alcohol
Polyamide (–CONH– link)	carboxylic acid + protonated amine (ammonium salt)	carboxylate salt + amine

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Q.44 [Long] What are photodegradable plastics? How does light degrade them?

Some functional groups can absorb more energy from sunlight in the form of UV light. Polymers with such functionalities can be degraded by light and are termed photodegradable. Some polymers with carbonyl groups (C=O) in their chains have been developed due to their enhanced photosorption. The absorption of UV light weakens the bonds within the chain and the forces between different chains of the polymer. This makes the polymer more biodegradable, accelerating its decomposition into less toxic compounds in the natural environment.

Q.45 [Short] What are the limitations of photodegradable plastics?

Photodegradable plastics cannot be used for a longer time, and recycling of such polymers is not efficient. Also, plastic is buried in landfills where no light is available for photodegradation.

Q.46 [Short] How is the biodegradability of polyesters and polyamides used in designing materials?

Because polyesters and polyamides can biodegrade, they do not stay in the environment forever like some other plastics; this helps to reduce the adverse effects of plastic pollution. Scientists and engineers use this knowledge to design materials that break down when they are no longer needed. For example, some medical implants and dissolvable stitches are made from polyesters that slowly break down in the body through hydrolysis.

26.6 POLYMERS IN ARTIFICIAL ORGANS

Q.47 [Long] What are artificial organs? What properties must polymers used in the body have?

In biomedical science, polymers are very important for creating artificial organs. These are man-made devices designed to replace or support the function of damaged or diseased body parts. Polymers used in the body need to be biocompatible, mechanically suitable, sterilizable, durable and tunable. They must not harm the body, cause allergies or be rejected by the immune system. Their strength, flexibility or elasticity can be adjusted to match the body part they replace (e.g. flexible for blood vessels, strong for bones).

Q.48 [Long] Outline the use of polymers in artificial organs.

Application	Polymers used	Key properties and functions
Artificial blood vessels (grafts)	Poly(tetrafluoroethylene) (PTFE) and poly(ethylene terephthalate) (PET)	Smooth surfaces to prevent blood clots; used to replace or bypass blocked vessels.
Artificial heart valves	Specialised polyurethanes and silicone	Highly flexible and durable; must be resistant to clotting.

Application	Polymers used	Key properties and functions
Artificial skin	Collagen (natural) and polylactic acid (PLA) (synthetic)	Acts as a scaffold for new skin growth; some are designed to dissolve as the wound heals.
Artificial joints (hip/knee)	Ultra-high molecular weight polyethylene (UHMWPE)	Provides a smooth, low-friction surface to allow movement and resist wear.
Artificial cornea	Poly(methyl methacrylate) (PMMA) and hydrogels	Must be perfectly clear (transparent) and biocompatible to restore vision.

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